

DSA Practical MCQs

1. (Time-Complexity of Pseudocode)

function first_code(n)

 i <- 2

 m <- n*n

 while i < m

 i <- i * 2

What is the running time Theta-notation?

a. Theta(n)

b. Theta(log₂ n)

c. Theta(n²)

d. Theta(2ⁿ)

Answer: b

2. (Multiset Operation)

A = {1,1,3,5,5,5,6,6}, B = {2,2,3,6,6,6,6}

What is A \ B (multiplicities of 1...6)?

a. 2,2,0,0,2,2

b. 2,0,0,0,2,2

c. 2,0,0,0,2,0

d. 2,2,0,0,2,0

Answer: c

3. (Sparse-Matrix Three-Row Representation)

Given matrix:

0 3 0

0 0 0

0 0 1

2 0 0

One column of its three-row storage is:

a. 3,2,1

b. 3,1,2

c. 2,1,3

d. 1,2,3

Answer: d

4. (Representation of a queue or stack)

A circular structure of capacity 5 stores elements in an array.

Initially: [1,2,3,-,-].

Operations:

enqueue(4), dequeue(), enqueue(5), enqueue(6), dequeue(), dequeue().

It may be used as a queue or as a stack. What happens?

a. [4,5,6,-,-]

b. [6,-,-,4,5]

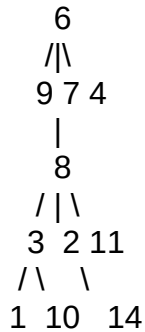
c. overflow

d. underflow

Answer: c

5. (Hierarchical-List for a General Tree)

Consider tree:



Which is its hierarchical list?

- a. 6(9,7(8(3(1,10),2,11(14))),4)
- b. 6(9,7(8(3(10,1),2,11(14))),4)
- c. 6(9,7(8(3(1)(10),2,11(14))),4)
- d. 6(9,7(8(3(1,10),2,11)(14)),4)

Answer: a

6. (Prefix or Postfix Expression)

Evaluate the expression (prefix or postfix):

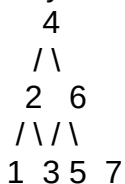
+ - * 5 2 3 / 6 2

- a. 5
- b. 7
- c. 10
- d. 13

Answer: c

7. (Search-Tree & Balance)

Binary tree:



Which applies?

- a. balanced search tree
- b. search tree, but not balanced
- c. balanced, but not a search tree
- d. neither balanced nor search tree

Answer: a

8a. (Graph Traversal: BFS)

Given the directed graph, start at 1 and perform BFS.

What is a valid BFS order of reachable nodes?

- a. 1, 3, 4, 6, 5
- b. 1, 3, 6, 4, 5
- c. 1, 2, 5, 3, 4
- d. 1, 4, 3, 6, 5

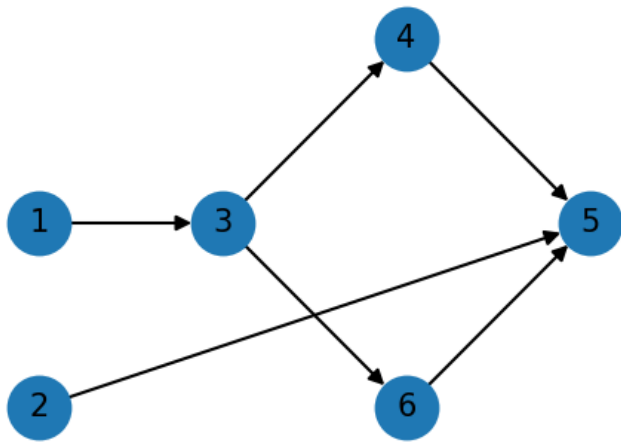
Answer: a

8b. (Graph Traversal: DFS)

Using the same graph, start at 1 and perform DFS.

What is a valid pre-order DFS traversal?

- a. 1, 3, 4, 5, 6
- b. 1, 3, 6, 5, 4
- c. 1, 2, 5, 3, 4
- d. 1, 4, 3, 6, 5



*** Questions 1, 2, 4, 6, 7 are very important for the exam ***

Good luck with the exam

- sadman